WORKED SOLUTION

STEP II 1997, Question 5

Pure • Complex Exponential Mapping

STEP II 1997, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q5

This question has a full guided treatment in *Beyond STEP*, Section 5.7, *The Complex Exponential as a Map*: https://beyondstep.keithmcnulty.org/ch05/complex-exponential.html

The key idea

The relation is just $z = \mathrm{e}^u \mathrm{e}^{\mathrm{i}v} = \mathrm{e}^w$, written in modulus–argument form: u controls the modulus of z and v controls its argument. Once you see that, every part is about polar coordinates. The one subtlety is that the argument is only defined up to multiples of 2π, so the map is many-to-one, and to get a one-to-one correspondence we must cut the u–v plane down to a horizontal strip of the right height, taking care over which edges to include.

Solution

Finding w when $z = \mathrm{i}\,\mathrm{e}$

Since $\mathrm{e}^u > 0$ and $|\cos v + \mathrm{i}\sin v| = 1$, the relation says that

$$|z| = \mathrm{e}^u, \qquad \arg z = v \pmod{2\pi}.$$

For $z = \mathrm{i}\,\mathrm{e}$ we have $|z| = \mathrm{e}$ and $\arg z = \frac{\pi}{2}$. So $\mathrm{e}^u = \mathrm{e}$, giving $u = 1$ (the real exponential is one-to-one), and $\cos v + \mathrm{i}\sin v = \mathrm{i}$, so $\cos v = 0$ and $\sin v = 1$, giving $v = \frac{\pi}{2} + 2k\pi$. Hence

$$w = 1 + \mathrm{i}\left(\tfrac{\pi}{2} + 2k\pi\right), \qquad k \in \mathbb{Z}.$$

There are infinitely many such w, spaced 2π apart on the vertical line $u = 1$.

The images of the lines $u =$ constant

On the line $u = c$, the point z has modulus e^c and argument v. As v runs over all real numbers, $\cos v + \mathrm{i}\sin v$ runs round the unit circle (infinitely often), so z traces the

$$\text{circle } x^2 + y^2 = \mathrm{e}^{2c}, \quad \text{centre } O, \text{ radius } \mathrm{e}^c.$$

Larger c gives a larger circle; $c = 0$ gives the unit circle; as $c \to -\infty$ the circles shrink towards O but never reach it, since $\mathrm{e}^c > 0$.

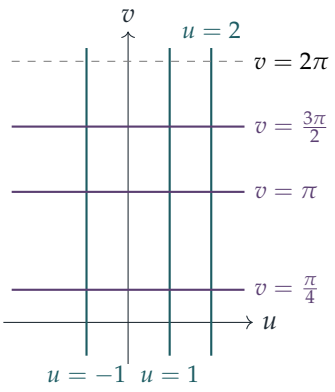
The images of the lines $v =$ constant

On the line $v = c$, we have $z = \mathrm{e}^u(\cos c + \mathrm{i}\sin c)$, a positive multiple of the fixed unit vector $(\cos c, \sin c)$. As u runs over $\mathbb{R}$, the multiple e^u takes every value in $(0, \infty)$ exactly once. So the image is the

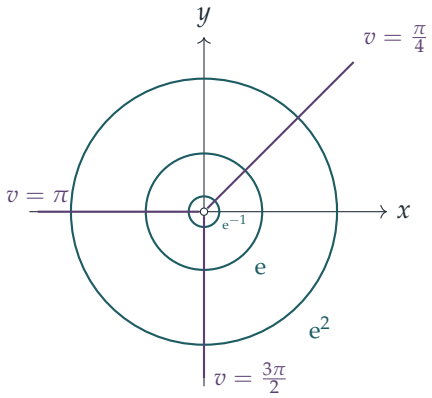
$$\text{half-line from } O \text{ at angle } c \text{ to the positive } x\text{-axis, with } O \text{ itself excluded.}$$

Lines $v = c$ and $v = c + 2k\pi$ give the same half-line. In Cartesian form, for example, $v = \frac{\pi}{4}$ gives $y = x$ with $x > 0$, and $v = \pi$ gives the negative x-axis.

The sketch shows the grid of lines in the u–v plane and its image: vertical lines become circles and horizontal lines become rays.



u–v plane



x–y plane

(The circles are not drawn to scale: the radii e^{-1}, e, e^2 grow very fast. The open circle marks O, which is never reached.)

Subsets in one-to-one correspondence with the first quadrant

A point z is in the open first quadrant exactly when $z \neq 0$ and it has an argument strictly between 0 and $\frac{\pi}{2}$. Take

$$W_1 = \left\{(u, v) : u \in \mathbb{R},\ 0 < v < \tfrac{\pi}{2}\right\}, \qquad W_2 = \left\{(u, v) : u \in \mathbb{R},\ 2\pi < v < \tfrac{5\pi}{2}\right\}.$$

Every point of W_1 maps into the first quadrant, since $e^u \cos v > 0$ and $e^u \sin v > 0$ there. Conversely, a point z of the first quadrant has $z = e^u(\cos v + i \sin v)$ exactly when $u = \ln|z|$ and $v = \theta + 2k\pi$, where $\theta \in (0, \frac{\pi}{2})$ is its principal argument. Precisely one of these values, $k = 0$, lies in $(0, \frac{\pi}{2})$, so z has exactly one preimage in W_1. Hence W_1 is in one-to-one correspondence with the quadrant. The same argument with $k = 1$ works for W_2, which is W_1 shifted up by 2π.

(Any strip $\{2k\pi < v < 2k\pi + \frac{\pi}{2}\}$ would do equally well.)

Subsets in one-to-one correspondence with $0 < |z| < 1$

Now we need $|z| = e^u < 1$, that is $u < 0$, and every argument must be hit exactly once, so v must run over an interval of length exactly 2π.

> **?** **Ask yourself** Should the strip include its top edge, its bottom edge, both or neither?

Exactly one. If we included both, say $0 \le v \le 2\pi$, then $(u, 0)$ and $(u, 2\pi)$ would both map to the same point e^u on the positive real axis, so the correspondence would fail to be one-to-one. If we included neither, the positive real axis would not be reached at all. Including just one edge makes every argument appear exactly once. So take

$$W_3 = \{(u, v) : u < 0,\ 0 \le v < 2\pi\}, \qquad W_4 = \{(u, v) : u < 0,\ -\pi < v \le \pi\}.$$

For W_3: if $0 < |z| < 1$ then $u = \ln|z| < 0$ is forced, and among the arguments $\theta + 2k\pi$ of z exactly one lies in $[0, 2\pi)$, because that interval has length 2π and contains only one of its endpoints. So each such z has exactly one preimage in W_3, and each point of W_3 maps to a z with $0 < e^u < 1$. For W_4 the same argument works with the interval $(-\pi, \pi]$, which corresponds to the principal argument. Both are half-planes $u < 0$ cut down to half-open strips of height 2π.

Checks

- Put $u = 1$, $v = \frac{\pi}{2} + 2\pi$: $z = e(\cos\frac{5\pi}{2} + i\sin\frac{5\pi}{2}) = ie$, as required.
- Numerically, for thousands of random (u, v) we found $|z| = e^u$ and $\arg z \equiv v$ to within 10^{-9}. For random points in each target set, we found exactly one preimage in each of $W_1, \ldots, W_4$. And the points $(-1, 0)$ and $(-1, 2\pi)$ both give $z = e^{-1} \approx 0.3679$, which confirms that a closed strip would double-count.

Common slips

- Giving only $w = 1 + \frac{\pi}{2}i$ and forgetting the $+2k\pi$.
- Including the origin in the images: e^u is never 0, so the circles never shrink to a point and the rays start at an excluded point.
- Using a closed strip $0 \le v \le 2\pi$ (or an open one $0 < v < 2\pi$) for W_3: the first

double-counts the positive real axis and the second misses it.

Beyond STEP

The map here is the complex exponential $w \mapsto \mathrm{e}^{w}$. It sends Cartesian grid lines to polar ones, and it is periodic with period $2\pi\mathrm{i}$. Choosing a strip on which it is one-to-one is exactly choosing a *branch of the complex logarithm*, its inverse. The edge you leave out is the *branch cut*, which explains why $\log z$ cannot be defined continuously on all of $\mathbb{C} \setminus \{0\}$. Tristan Needham's *Visual Complex Analysis* has beautiful pictures of this mapping and of conformal maps in general.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.